

RESEARCH ARTICLE

A Device-Physics-Informed Artificial Neural Network Approach for Thermal-Aware I - V and C - V Modeling of GAA FETs

YUN TAI^{1,2}, YIMING LI^{1,2,3,4,5,6,7}, (Senior Member, IEEE),
AND MIN-HUI CHUANG^{1,3}, (Student Member, IEEE)

¹Parallel and Scientific Computing Laboratory, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

²Institute of Intelligent Bioelectrical Engineering, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

³Institute of Communications Engineering, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

⁴Department of Electronics and Electrical Engineering, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

⁵Department of Microelectronics, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

⁶Institute of Pioneer Semiconductor Innovation, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

⁷Institute of Artificial Intelligence Innovation, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

Corresponding author: Yiming Li (ymli@nycu.edu.tw)

This work was supported in part by the National Science and Technology Council (NSTC), Taiwan, under Grant NSTC 114-2221-E-A49-192, Grant NSTC 114-2218-E-006-022-MBK, Grant NSTC 113-2221-E-A49-094, and Grant NSTC 113-2218-E-006-019-MBK.

ABSTRACT This work introduces a device-physics-informed neural network framework for simultaneous modeling of thermal-aware I - V and C - V characteristics of gate-all-around (GAA) field-effect transistors (FETs). The approach integrates the Grove–Frohnman I - V and Meyer C - V analytical expressions within artificial neural networks (ANNs), enforcing a shared temperature-dependent threshold voltage and embedding temperature effects in both ANN parameters and analytical expressions. A Gummel symmetry test is applied across temperatures to ensure current symmetry. Experimental validation demonstrates that the proposed method maintains high predictive accuracy even with limited training data, and generalizes well beyond the training conditions, while achieving significantly faster circuit simulation speed and reduced modeling effort compared to the well-known BSIM-CMG model. These results highlight the framework’s potential as a scalable, physically interpretable, and manufacturing-aware solution for advanced transistor modeling and circuit-level simulations.

INDEX TERMS Device-physics-informed, neural network, GAA, nanosheet, complementary, FETs, temperature effects, I - V and C - V models, device model, circuit simulation.

I. INTRODUCTION

Machine learning algorithms have become vital in semiconductor device research, enabling advances in compact modeling [1], parameter extraction [2], variability assessment [3], [4], [5], and performance tuning [6], [7]. Among these, compact models serve a critical role by bridging IC design and fabrication. Traditional models, such as the Berkeley Short-channel IGFET Model (BSIM) [8], are grounded in physical principles and consist of numerous interpretable parameters. However, as device dimensions

shrink [9], [10], [11], [12], [13], they require increasingly complex compact models to account for advanced physical phenomena, leading to higher computational costs and more challenging parameter extraction. In response, deep learning (DL) approaches have gained traction as flexible, data-driven alternatives capable of modeling highly nonlinear device behavior with improved accuracy and efficiency [14], [15], [16], [17]. In 2021, Q. Yang et al. [18] proposed a multigradient ANNs specifically for GAA silicon FETs. Their approach integrated gradients into the backpropagation process and applied gradient constraints within the ANN, capturing higher-order derivatives of device characteristics and eliminating the need for manually derived gradient

The associate editor coordinating the review of this manuscript and approving it for publication was Bing Li¹.

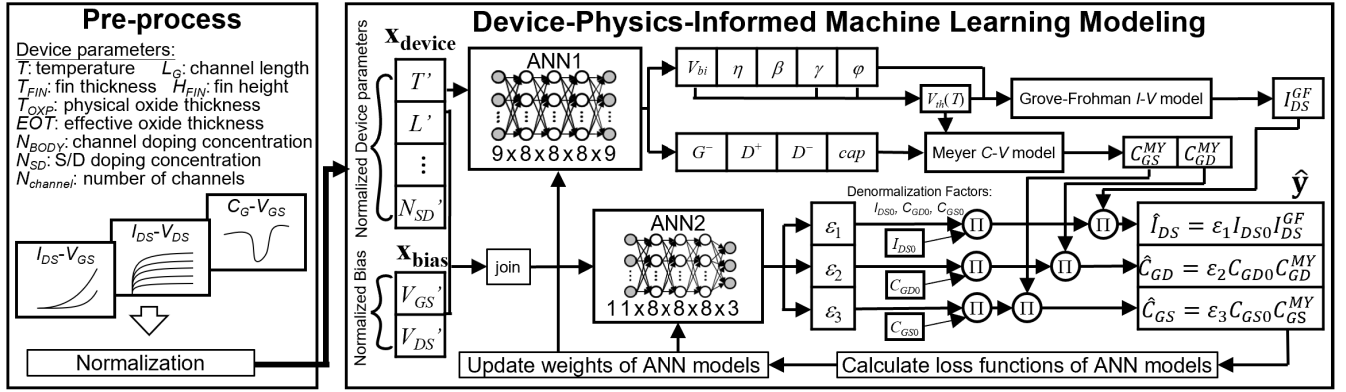


FIGURE 1. The entire training process of the device-physics-informed model is structured into two distinct artificial neural networks, ANN1 and ANN2. ANN1 is responsible for taking all device parameters and T as inputs and generating nine coefficients that are used in the GF and MY models with a shared $V_{th}(T)$. These models, when combined with temperature effects and other contributing factors, yield the output current I_{DS}^{GF} as well as two capacitances C_{GS}^{MY} , C_{GD}^{MY} . ANN2, on the other hand, processes both device parameters and bias conditions as inputs and produces three amendment factors that refine the model's accuracy in the simulation process. For clarity and to maintain the flow of presentation, detailed definitions of symbols and their physical meanings are provided in Appendix.

formulas in the loss function. C.-T. Tung et al. [19], [20] further refined capacitance modeling by including higher-order derivative constraints for capacitance in the ANN loss function, which yielded improved accuracy in capacitance simulations. Nonetheless, errors persisted at the boundary points of I - V and C - V curves due to limitations in capturing precise derivative behavior under boundary conditions. To address these boundary-related inaccuracies, M. Li et al. [21] proposed a novel physics-inspired neural network (Pi-NN) model, embedding fundamental device physics within the ANN framework. The Pi-NN model demonstrates smooth, accurate, and computationally efficient device simulations and performs effectively with discrete data—a crucial advantage for real-world novel devices where measurements are often discrete rather than continuous. This physics-based approach thus enables the Pi-NN model to provide robust and accurate simulations for emerging device technologies. Despite these advancements, temperature effects have not been adequately considered. Temperature is a crucial factor that affects the operation of chips [22], [23], [24], as elevated temperature (T) not only increases carrier concentration but also amplifies leakage currents, potentially impacting both device performance and reliability. Thus, temperature effects must be incorporated into ANN-based compact models. J. Wang et al. [25] incorporated T as a parameter in their ANN model, attempting to account for temperature effects in device simulations. However, due to the lack of a strong physical basis within their ANN framework, their model exhibits limitations similar to those previously discussed.

In this article, we propose a device-physics-informed ML modeling technique that combines the strengths of the Grove–Frohm (GF) I - V [26] and Meyer (MY) C - V [27] models. The framework further advances two ANN models, ANN1 and ANN2, both are designed for a specific modeling task, inspired by our recent work in [28]. By using a shared threshold voltage, ANN1 takes all device parameters (eight

TABLE 1. The hyperparameters of ANN1 and ANN2.

Settings	ANN1	ANN2
Hidden layers	3	3
Neurons	8, 8, 8	8, 8, 8
Epochs	3000	3000
Learning rate	0.01	0.01
Scheduler	StepLR	StepLR
Activation function	tanh	tanh
Loss function	RMSE	RMSE
Input dimension	(16, 9)	(16, 11)
Output dimension	(16, 9)	(16, 3)
Optimizer	SGD	SGD

parameters in this study) and temperature as input and outputs nine temperature-dependent coefficients for the GF and MY models. ANN2 takes device, temperature, and bias conditions (total number of parameters: eleven) as inputs and outputs three amendment factors for I - V and C - V curves. Our approach bridges data-driven modeling with physical insights across a wide range of operating conditions, temperatures, and device geometries.

This paper is organized as follows. In Section II, we state the temperature-dependent models and the modeling strategy. In Section III, we discuss the achieved results in device and circuit levels. Section IV draws conclusions and suggests future works.

II. TEMPERATURE DEPENDENT I - V AND C - V MODELS

Fig. 1 is a detailed overview of the device-physics-informed ML modeling technique. First, we deal with preprocessing device parameters to ensure data quality and consistency. We compress the targeted I - V and C - V curves including all input parameters to the interval $[0, 1]$ linearly. This step is crucial for the stability of the ANN models [28]. Next, ANN1 is trained to map geometry- and temperature-dependent

Algorithm 1 Algorithm for Training ANN1

```

1: Input:  $D = \{(x_i, y_i)\}_{i=1}^N$ ,  $x_i = [T, L_G, T_{FIN}, H_{FIN}, T_{OXP}, EOT, N_{BODY}, N_{SD}, N_{channel}] \in \mathbb{R}^9$ ,  $y_i = [I_{DS}, C_{GD}, C_{GS}]$ 
2: Output:  $\theta_1$  to predict GF/MY coefficients
3: Randomly initialize  $\theta_1$ 
4: for epoch 1 to 3000 do
5:   for each batch  $\{x_j\}_{j=1}^{16}$  from  $D$  do
6:     for each  $x_j$  in the batch do
7:        $h_0 \leftarrow x_j$ 
8:       for  $l = 1$  to 4 do
9:          $h_l \leftarrow \tanh(W_l \cdot h_{l-1} + b_l)$ 
10:      end for
11:       $z_j \leftarrow \sigma(W_{out} \cdot h_4 + b_{out})$ 
12:      Use  $z_j$  in GF and MY models to compute  $\hat{y}_j = [\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}]$ 
13:       $\mathcal{L}_j \leftarrow RMSE([\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}], [I_{DS}, C_{GD}, C_{GS}])$ 
14:    end for
15:    Update  $\theta_1$  using SGD optimizer with StepLR
16:  end for
17: end for
18: return  $\theta_1$ 

```

Algorithm 2 Algorithm for Training ANN2

```

1: Input:  $D = \{(x_i, y_i)\}_{i=1}^N$ ,  $x_i = [T, L_G, T_{FIN}, H_{FIN}, T_{OXP}, EOT, N_{BODY}, N_{SD}, N_{channel}, V_{GS}, V_{DS}] \in \mathbb{R}^{11}$ 
2: Output:  $\theta_2$  to generate three amendment factors
3: Randomly initialize  $\theta_2$ 
4: for epoch 1 to 3000 do
5:   for each batch  $\{x_j\}_{j=1}^{16}$  from  $D$  do
6:     for each  $x_j$  in the batch do
7:        $h_0 \leftarrow x_j$ 
8:       for  $l = 1$  to 4 do
9:          $h_l \leftarrow \tanh(W_l \cdot h_{l-1} + b_l)$ 
10:      end for
11:       $o_j \leftarrow \sigma(W_{out} \cdot h_4 + b_{out})$ 
12:      Use  $o_j$  to compute  $\hat{y}_j = [\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}]$  via correction formulation
13:       $\mathcal{L}_j \leftarrow RMSE([\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}], [I_{DS}, C_{GD}, C_{GS}])$ 
14:    end for
15:    Update  $\theta_2$  using SGD optimizer with StepLR
16:  end for
17: end for
18: return  $\theta_2$ 

```

device parameters to physically meaningful coefficients used in the GF current model and the MY capacitance model. These coefficients preserve the analytical structure of the original physics-based formulations while allowing temperature dependencies to be learned directly from data. To further address bias effects and modeling inaccuracies not captured by the analytical expressions, ANN2 is introduced to generate

multiplicative correction factors. These factors compensate for short-channel effects, bias-dependent nonidealities, and scalability limitations inherent in classical compact models.

The architecture of the ANN models was selected through empirical hyperparameter testing, where the number of hidden layers and neurons per layer were varied from 2 to 5 and from 4 to 16, respectively. The adopted structure, consisting of three hidden layers with eight neurons each, provides a good balance between modeling accuracy and computational efficiency. Compared with direct interpolation of discrete data, the ANN provides a continuous and differentiable representation, reducing numerical artifacts and improving robustness under unseen bias or temperature conditions, which is essential for stable SPICE-compatible circuit simulations. Table 1 lists the hyperparameters used in training ANN1 and ANN2, where the stochastic gradient descent (SGD) optimizer is adopted.

The models are trained on 100 devices of varying sizes across a wide range of operating conditions to ensure robustness and generalizability. This training stage serves as a pretraining process to learn a general mapping between device parameters, temperature, and the corresponding physical model parameters. For specific device applications (e.g., nanosheet-FETs (NS-FETs) or complementary FETs (CFETs)), the pretrained model is further fine-tuned using a smaller dataset tailored to the target device. A Sigmoid activation function is applied after the output layer of ANN1 to ensure that the range of output values remains between 0 and 1. As shown in Algorithm 1, the input vector $x \in \mathbb{R}^9$ consists of eight device parameters and temperature. The parameter set θ_1 comprises all weights and biases of ANN1. The outputs $z = [V_{bi}, \eta, \beta, \gamma, \varphi, G^-, D^+, D^-, cap]$ (see Appendix for their explanations) of ANN1 are subsequently used as physically meaningful parameters in the analytical current and capacitance formulations based on the GF and MY models, respectively.

Notably, the GF model, derived from fundamental semiconductor device physics, is widely implemented in SPICE simulators as the LEVEL-2 MOSFET model [28]. It incorporates bulk charge effects, mobility degradation under high vertical electric fields, carrier velocity saturation, and threshold voltage variation [22]. Despite its physical rigor, the GF model exhibits inherent limitations when applied to deeply scaled short-channel devices, particularly due to the absence of key short-channel effects and limited scalability. For capacitance modeling, the MY model is adopted as a foundational MOSFET capacitance formulation that has been extensively used for circuit-level simulation. Its primary benefits are its simplicity and ease of parameterization. Nevertheless, the MY model suffers from notable drawbacks, particularly its reduced accuracy for short-channel devices and a fundamental issue of charge non-conservation. These shortcomings motivate the integration of data-driven corrections within a physics-informed learning framework.

In this framework, ANN1 predicts a set of physically interpretable parameters, including V_{bi} , η , β , γ , and φ ,

which explicitly appear in the GF current formulation [26], [28]. Although these parameters have implicitly depended on temperature via the ANN1, the threshold voltage is further modeled to explicitly capture its temperature dependence as

$$V_{th}(T) = \left[V_{bi} + \frac{3}{2} \gamma \sqrt{\varphi \times (t_{rel})} + \frac{\gamma}{\sqrt{\varphi \times (t_{rel})}} \right] \times t_{rel}^{-1.5}, \quad (1)$$

where $t_{rel} = T/T_{ref}$ and T_{ref} is the room temperature in Kelvin. The calculated $V_{th}(T)$ is further applied to evaluate gate capacitances: C_{GS}^{MY} and C_{GD}^{MY} using the expressions in the MY model [27]. For $V_{DS} < 0.1$ V and $V_{GS} < V_{th} + 0.1$ V,

$$C_{GS}^{MY} = cap \times G^- \times D^- \times \left\{ \frac{V_{GS} - V_{th}}{0.1} \times \left[1 - \left(\frac{0.1 - V_{DS}}{0.2 - V_{DS}} \right)^2 - D^- \right] + D^- \right\} \quad (2)$$

and

$$C_{GD}^{MY} = cap \times G^- \times D^+ \times \left\{ D^+ + \frac{V_{GS} - V_{th}}{0.1} \times \max \left[0.1 - \left(\frac{0.1 - V_{DS}}{0.2 - V_{DS}} \right)^2 - D^+, 0 \right] \right\} \quad (3)$$

For $V_{DS} < 0.1$ V and $V_{GS} \geq V_{th} + 0.1$ V,

$$C_{GS}^{MY} = cap \times G^- \times D^- \times \left\{ 1 - \left[\frac{V_{GS} - V_{th} - V_{DS}}{2 \times (V_{GS} - V_{th}) - V_{DS}} \right]^2 \right\} \quad (4)$$

and

$$C_{GD}^{MY} = cap \times G^- \times D^+ \times \max \left\{ D^+, 1 - \left[\frac{V_{GS} - V_{th} - V_{DS}}{2 \times (V_{GS} - V_{th}) - V_{DS}} \right]^2 \right\} \quad (5)$$

For $V_{DS} \geq 0.1$ V,

$$C_{GS}^{MY} = cap \times G^- \times \left\{ 1 - \left[\frac{V_{GS} - V_{th} - V_{DS}}{2 \times (V_{GS} - V_{th}) - V_{DS}} \right]^2 \right\} \quad (6)$$

and

$$C_{GD}^{MY} = cap \times G^- \times D^+ \times \max \left\{ D^+, 1 - \left[\frac{V_{GS} - V_{th} - V_{DS}}{2 \times (V_{GS} - V_{th}) - V_{DS}} \right]^2 \right\} \quad (7)$$

Notably, cap is the gate oxide capacitance associated with the effective channel area; G^- , D^- , and D^+ are smoothing factors.

Meanwhile, as specified in Algorithm 2, ANN2 takes device geometry, temperature, and bias conditions as inputs $x \in \mathbb{R}^{11}$ and outputs the parameters ϕ_1 , ϕ_2 , and ϕ_3 . θ_2 includes all weights and biases for ANN2. The correction factors are defined as $\varepsilon_1 = 10^{\phi_1}$, $\varepsilon_2 = 10^{\phi_2}$, and $\varepsilon_3 = 10^{\phi_3}$. The final outputs $\hat{I}_{DS}$, $\hat{C}_{GD}$, and $\hat{C}_{GS}$ are calculated by $\hat{I}_{DS} = \varepsilon_1 I_{DS0} I_{DS}^{GF}$, $\hat{C}_{GD} = \varepsilon_2 C_{GD0} C_{GD}^{MY}$, and $\hat{C}_{GS} = \varepsilon_3 C_{GS0} C_{GS}^{MY}$,

Algorithm 3 Algorithm for Training Full Model

```

1: Input:  $D = \{(x_i, y_i)\}_{i=1}^N$ ,  $x_i \in \mathbb{R}^{11}$ 
2: Output:  $\theta_1$  and  $\theta_2$  are the trained parameters of ANN1 and ANN2, respectively
3: Randomly initialize  $\theta_1, \theta_2$ 
4: for epoch 1 to 3000 do
5:   for each batch  $\{x_j\}_{j=1}^{16}$  from  $D$  do
6:     for each  $x_j$  in the batch do
7:        $z_j \leftarrow \text{ANN1}([x_j, 1:9]; \theta_1)$ 
8:        $T \leftarrow x_{j,1}$ 
9:        $t_{rel} \leftarrow T/T_{ref}$ 
10:       $V_{bi} \leftarrow z_{j,1}$ 
11:       $\eta \leftarrow z_{j,2}$ 
12:       $\beta \leftarrow z_{j,3}$ 
13:       $\gamma \leftarrow z_{j,4}$ 
14:       $\varphi \leftarrow z_{j,5}$ 
15:      Compute  $V_{th}(T)$  by (1)
16:       $I_{DS}^{GF} \leftarrow GF([z_{j,1:5}], V_{th})$ 
17:       $G^- \leftarrow z_{j,6}$ 
18:       $D^+ \leftarrow z_{j,7}$ 
19:       $D^- \leftarrow z_{j,8}$ 
20:       $cap \leftarrow z_{j,9}$ 
21:       $(C_{GS}^{MY}, C_{GD}^{MY}) \leftarrow MY([z_{j,6:9}], V_{th})$ 
22:       $\phi_j \leftarrow \text{ANN2}(x_j; \theta_2)$ 
23:       $\varepsilon_j \leftarrow 10^{\phi_j}$ 
24:       $I_{DS} \leftarrow \varepsilon_{j,1} \times I_{DS0} \times I_{DS}^{GF}$ 
25:       $C_{GD} \leftarrow \varepsilon_{j,2} \times C_{GD0} \times C_{GD}^{MY}$ 
26:       $C_{GS} \leftarrow \varepsilon_{j,3} \times C_{GS0} \times C_{GS}^{MY}$ 
27:       $\mathcal{L}_j \leftarrow \text{RMSE}([\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}], [I_{DS}, C_{GD}, C_{GS}])$ 
28:    end for
29:    Update  $\theta_1, \theta_2$  using SGD optimizer with StepLR
30:  end for
31: end for
32: return  $\theta_1, \theta_2$ 

```

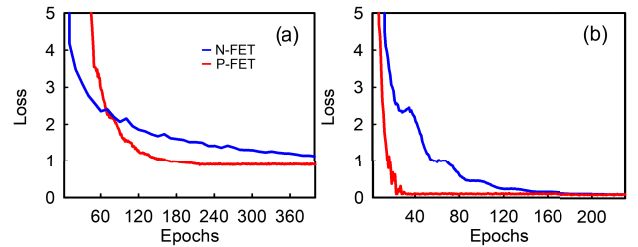


FIGURE 2. The training convergence of GAA (a) NS-FETs and (b) CFETs.

respectively, where I_{DS0} , C_{GD0} , and C_{GS0} are dimensionless denormalization factors. This strategy is applied to a pre-normalized database of the targeted I - V and C - V curves, helping maintain model accuracy. Both the ANN1 and ANN2 models are trained via backpropagation, a gradient-based optimization algorithm, to minimize the loss function defined as the root mean squared error (RMSE) between predicted $\hat{y} = [\hat{I}_{DS}, \hat{C}_{GD}, \hat{C}_{GS}]$ and actual $y = [I_{DS}, C_{GD}, C_{GS}]$.

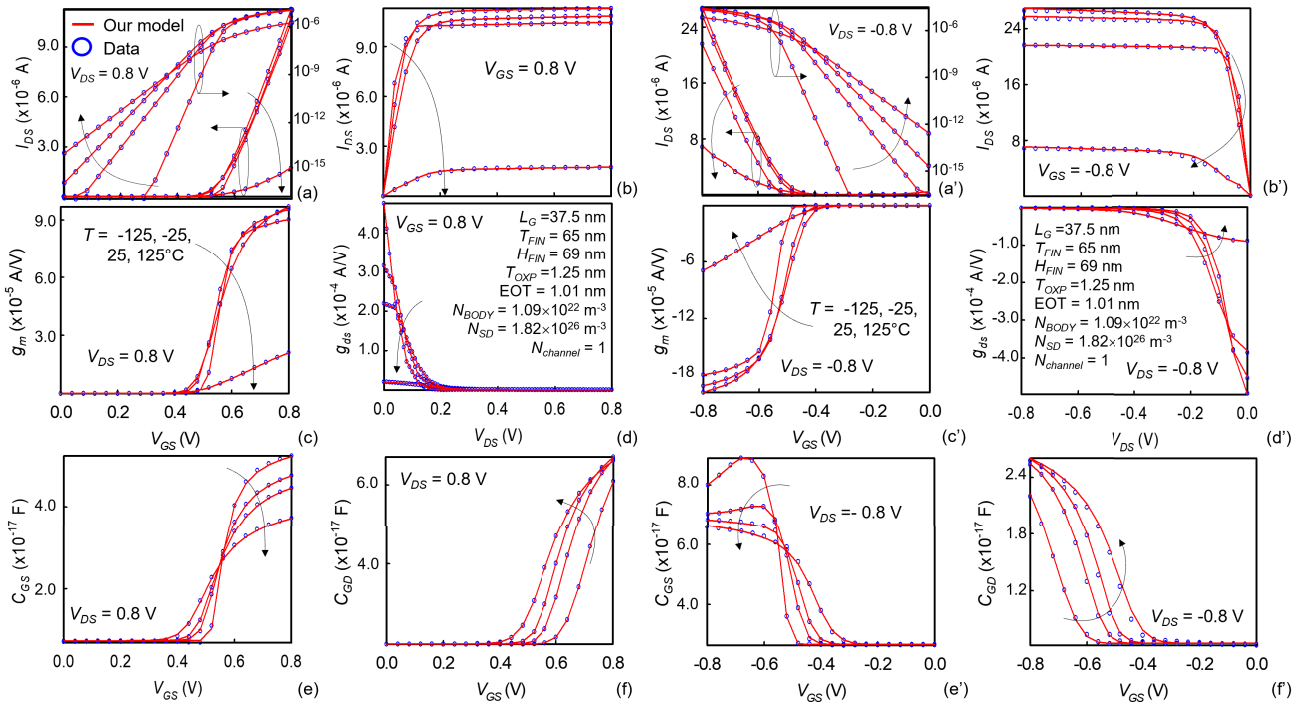


FIGURE 3. The prediction of the (a)-(f) N- and (a')-(f') P-FET simulation results across the voltage range of 0 to 0.8 V using our model at different temperatures. Despite the model is not trained for these specific conditions, it is able to accurately simulate the device behavior.

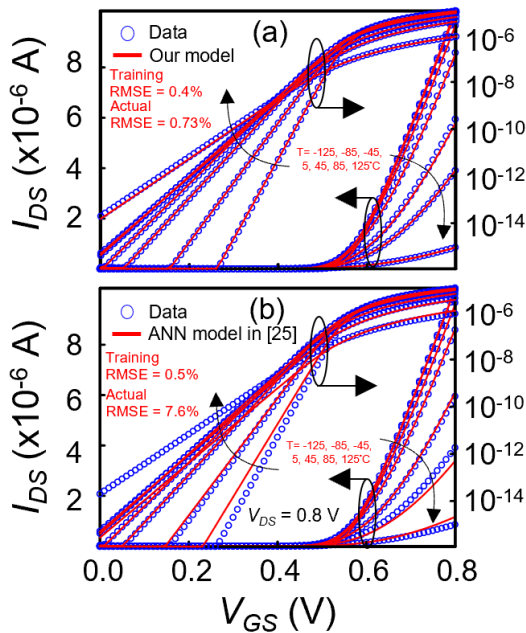


FIGURE 4. Comparison between (a) this work and (b) the ANN model in [25] for GAA NS N-FETs across different temperatures. Notably, the curves at -125°C and 125°C extend beyond the training range.

Algorithm 3 summarizes the complete training procedure. The network is trained for 3000 epochs, with parameters iteratively updated at each step. For completeness, the

key symbols and their physical meanings used in the GF/MY formulations and ANN framework are summarized in Appendix. This systematic training ensures that the ANN effectively learns the complex, nonlinear relationships governing the device characteristics, enabling accurate and physically consistent predictions.

III. RESULTS AND DISCUSSION

In this section, we validate and compare our trained models via various tests in device and circuit levels. The temperature-dependent I - V and C - V models are trained for all biases, where T varies from -100°C to 100°C . The input ranges for V_{GS} and V_{DS} are between 0 and 0.8 V, with steps of 0.04 V. The convergence of the ANN models during training is monitored using the loss function of RMSE. We randomly generated 30 different devices, with device parameters sampled within predefined ranges to ensure model generalization: L_G varies between 10 and 80 nm, while the T_{FIN} and H_{FIN} are derived as fractions from 0.4 to 0.7 of L_G . T_{OXP} , EOT , N_{BODY} , and N_{SD} are set to their default values of 1.2 nm, 1 nm, $1 \times 10^{22} / \text{m}^3$, $2 \times 10^{26} / \text{m}^3$, respectively, with a $\pm 10\%$ variation in BSIM-Common Multi-Gate (CMG) to reflect typical process-induced variations in nowadays technologies [29]. This corresponds to T_{OXP} (1.08–1.32 nm), EOT (0.9–1.1 nm), N_{BODY} (0.9 – $1.1 \times 10^{22} / \text{m}^3$), and N_{SD} (1.8 – $2.2 \times 10^{26} / \text{m}^3$). $N_{channel}$ is chosen from one to three. These numeric ranges are designed to capture device variations while ensuring sufficient diversity in training data.

Figs. 2(a) and (b) show the training convergence of the ANN models of N-/P-FETs. In Fig. 2(a), for the GAA NS-FETs, the model of P-FETs even has a lower initial loss because it was trained following the pre-trained results of N-FETs. To highlight the concept of transfer learning [30], since device physics is incorporated during the training process, we can inherit the training coefficients of the ANN model to accelerate convergence. N-/P-FETs possess different current and capacitance values, but they share similar parameters that facilitate rapid convergence. Fig. 2(a) shows that the P-FET model converges more rapidly than the N-FET model, owing to the fact that it was initialized using the pre-trained N-FET model rather than random coefficients. This indicates that having a pre-trained model can substantially reduce the training time for different devices, which is highly beneficial for further device training. We have applied the trained I - V and C - V models of GAA NS N-/P-FETs to train CFETs. Fig. 2(b) illustrates the training convergence for N-/P-FETs of GAA CFETs; notably, the N-FET converges faster than the training without a trained model of GAA NS N-FET. For the P-FET of GAA CFETs, we similarly leveraged the trained ANN model of the GAA NS P-FET, resulting in an even faster convergence (about 30 epochs).

A. DEVICE CHARACTERISTICS PREDICTION

We now model the I - V and C - V curves of GAA NS N-/P-FETs for all T . For model evaluation, we statistically generate device parameters outside the training data: $L_G = 37.5$ nm, $T_{FIN} = 65$ nm, $H_{FIN} = 69$ nm, $T_{OXP} = 1.25$ nm, $EOT = 1.01$ nm, $N_{BODY} = 1.09 \times 10^{22}$ m $^{-3}$, $N_{SD} = 1.82 \times 10^{26}$ m $^{-3}$, and $N_{channel} = 1$. These parameters are paired with varying temperatures and entered into the ANN model. Figs. 3(a)-(f) show I_{DS} - V_{GS} in linear and log scales, I_{DS} - V_{DS} , g_m - V_{GS} , g_{ds} - V_{DS} , C_{GS} - V_{GS} , and C_{GD} - V_{GS} of N-FETs, respectively, while Figs. 3(a')-(f') show the results of P-FETs. The model performs well even at -125°C and 125°C , where T is beyond the training range, because the temperature-dependent equations are introduced. We do emphasize that temperature is not only an input variable of ANNs, but also serves as a factor in the device equations, thereby enhancing its impact on device characteristics. Fig. 4 further compares our ANN model with the ANN model by J. Wang et al. [25], which only incorporates temperature as an input to their ANN. The blue circles are data generated from BSIM-CMG, while the red lines are results of ANNs. They are trained with the same number of epochs and dataset for $T = -85^\circ\text{C}$, -45°C , 5°C , 45°C , and 85°C . Within the training range of T , both models achieve low training RMSE values of 0.4% and 0.5%. However, for the predicted I_{DS} - V_{GS} curves at $T = -125^\circ\text{C}$ and 125°C , outside the training range, our model still maintains a low RMSE of 0.7%, whereas the RMSE of ANN model in [25] increases to 7.6%.

Fig. 5 shows evaluations of our models across different L_G : 16, 20, 50, and 90 nm; it shows the model's robustness when predicting 25% beyond the training voltage range. Notably,

as L_G increases from 16, 20, 50, to 90 nm, the drain-induced barrier lowering (DIBL) decrease gradually from 97.56, 53.84, 8.46, to 5.04 mV/V, respectively, while our model's results are 96.51, 53.54, 8.37, and 4.96 mV/V, with RMSE below 0.6%, indicating that the model effectively captures the behavior of devices across a wide range of dimensions. Unlike Fig. 3, where the training range was from 0 to 0.8 V with a 0.04 V step, Fig. 5 extends the range to 1 V and reduces the step size to 0.01 V. Despite this finer resolution and extended voltage range, the model still maintains a low RMSE of 0.53% across all I - V and C - V curves; in particular, for high bias conditions. The aforementioned results confirm the model's adaptability and high accuracy for all verification scenarios. Not shown here, we also examine the model prediction for the device dimension, voltage, and temperature beyond the training data, achieving an RMSE within 1%.

Building on the demonstrated predictive accuracy, we further evaluate the physical interpretability of the proposed model through a sensitivity analysis of selected parameters, including N_{SD} , EOT , and T , for the device, as shown in Fig. 3. In this analysis, all parameters except the one under investigation are fixed at their nominal values, while the parameter of interest is statistically sampled from a normal distribution with 300 realizations. The impact of parameter variations is quantified using the relative standard deviation, defined as $\sigma/\mu \times 100\%$, where μ and σ denote the averaged value and the standard deviation, respectively. Temperature variations exhibit the most pronounced impact, as shown in Fig. 6, yielding relative standard deviations of 17.5% for I_{on} ($\sigma_{I_{on}} = 0.919\mu\text{A}$, $\mu_{I_{on}} = 5.24\mu\text{A}$) and 4.43% for I_{off} ($\sigma_{I_{off}} = 0.0209\text{fA}$, $\mu_{I_{off}} = 0.472\text{fA}$). This observation confirms that temperature is a dominant factor influencing device operation, consistent with established physical understanding [22]. In comparison, variations in EOT , not shown here, result in smaller dispersions of 4.6% for I_{on} and 0.984% for I_{off} . Variations in N_{SD} lead to relative standard deviations below 1% for both metrics. Overall, this sensitivity analysis demonstrates that the proposed physics-informed ANN model accurately captures the continuous dependency of model parameters for the explored devices.

To further validate the model's capability in capturing higher-order device behavior, we performed the Gummel symmetry test (GST), which evaluates current symmetry and higher-order derivatives in compact models [31]. The GST is verified for the proposed ANN model with $T = 125^\circ\text{C}$, 25°C , and -25°C at $V_{GS} = 0.8$ V. Figs. 7(a)-(d) show the model's I_{DS} response up to the third-order derivatives; Fig. 7(b) is the first derivative, i.e., the transconductance, while the second and third derivatives reveal higher-order nonlinearities, as shown in Figs. 7(c) and (d).

Besides examining the trend of outputs of ANN1 vs. L_G [28], Figs. 8(a) and (b) further show the normalized coefficients vs. temperature for the GF and MY models. To verify temperature-dependent trends, a selected device is with $L_G = 38$ nm and $T_{FIN} = H_{FIN} = 21$ nm, which is out of the training data. The results indicate that three

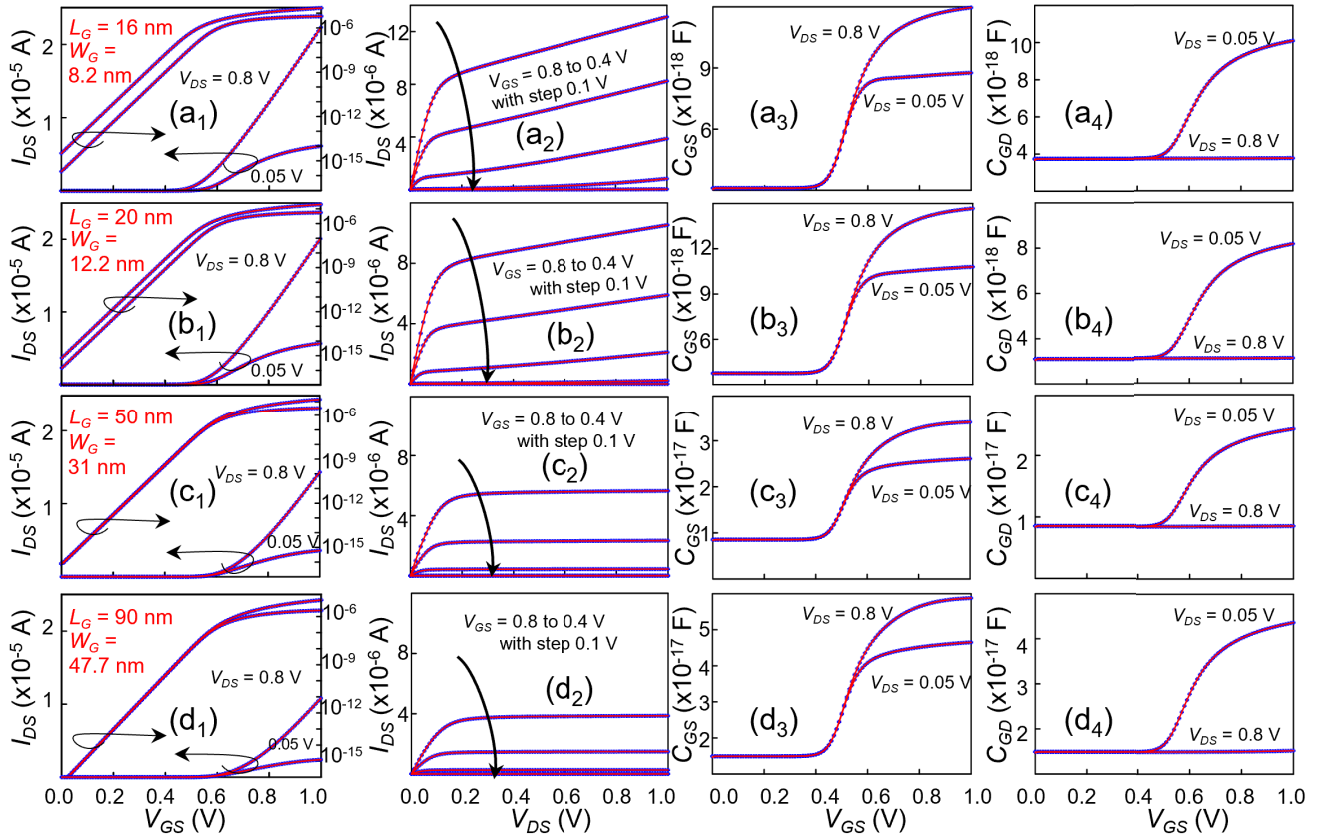


FIGURE 5. The predicted I_{DS} - V_{GS} , I_{DS} - V_{DS} , C_{GS} , and C_{GD} of the GAA NS N-FETs with L_G = (a1)-(a4) 16, (b1)-(b4) 20, (c1)-(c4) 50, and (d1)-(d4) 90 nm, respectively, by using the trained ANN models, where blue symbols are data and red lines are modeled results.

parameters β , φ directly from the ANN1 outputs and V_{th} exhibit significant changes with temperature. In contrast, the others remain unaffected by temperature. They are consistent with the physical trends in GF and MY models. Fig. 8(c) lists the numeric range of these parameters and Fig. 8(d) compares the I_{DS} - T characteristics predicted by our ANN model against those obtained from the BSIM-CMG. Despite being trained at only five discrete temperature points, our model achieves high accuracy across the entire temperature range, demonstrating the effectiveness of incorporating temperature-dependent physical parameters into the ANN framework. These results also validate the robustness of the proposed model's predictions across different temperatures. Compared to our previous work [32], as summarized in Table 2, we have significantly expanded the set of input parameters by further incorporating L_G , T_{FIN} , T_{OXP} , and N_{SD} . This enhancement enables the model to account for a broader range of device characteristics with respect to various device dimensions. The temperature range is extended from -25°C to 105°C to a broader -100°C to 100°C , representing a 53% increase. The optimized data usage reduces the dataset size from 200,000 to 26,460 samples, corresponding to only about 13% of the original data, while maintaining comparable accuracy. This reduction is primarily attributed to incorporating temperature-adjusted parameters

TABLE 2. Comparison between this work and our previous work.

Comparison	This work	[32]
Input Parameters	V_{DS} , V_{GS} , L_G , T , T_{FIN} , H_{FIN} , T_{OXP} , EOT , N_{BODY} , N_{SD} , $N_{channel}$	V_{DS} , V_{GS} , T , H_{FIN} , EOT , N_{BODY} , $PHIG$
Output Parameters	I_{DS} , C_{GS} , C_{GD}	I_{DS}
Data requirement	26460	200000
Temperature Range	$-100^\circ\text{C} - 100^\circ\text{C}$	$-25^\circ\text{C} - 105^\circ\text{C}$
Convergence Speed (epoch)	672	3127
Training Time (hr)	3.17	21.86
Accuracy	99.3%	98.9%
I_{DS} Predictability	High	High
C_{GS} and C_{GD} Predictability	High	None
Optimizer	SGD	SGD

into the GF model. The optimized training strategy effectively minimizes the required data volume and training time without sacrificing predictive accuracy across different devices and temperatures.

We further extend the proposed framework to model I_{DS} - V_{DS} , I_{DS} - V_{GS} , C_{GS} - V_{GS} and C_{GD} - V_{GS} of 16-nm GAA Si NS CFET across various temperatures. The simulated device

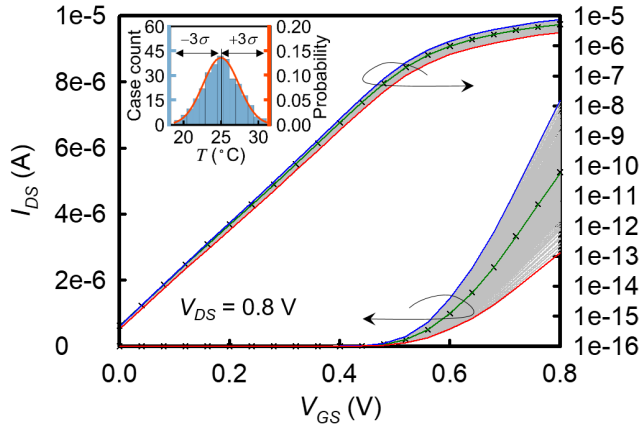


FIGURE 6. Sensitivity analysis of the I_{DS} - V_{GS} characteristics of GAA NS N-FETs with respect to temperature T . The gray curves represent the responses of 300 randomly sampled devices, while the black crosses denote the nominal device at 25°C. The green curves indicate the ensemble-averaged response, and the blue and red curves correspond to the lowest and highest sampled temperatures, respectively. The insets show the normal distribution of the sampled temperature, with a mean of 25°C and a standard deviation of 2.5°C. All sampled cases fall within the range of -3σ to $+3\sigma$.

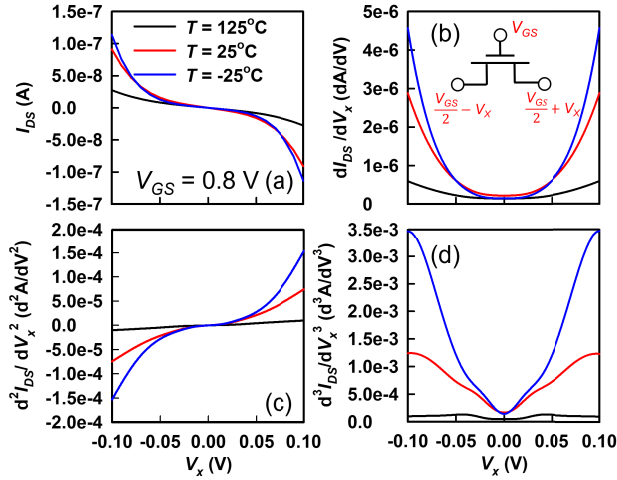


FIGURE 7. (a)-(d) The Gummel symmetry tests of the reported I - V model at $V_{GS} = 0.8$ V with $T = 125^\circ\text{C}$, 25°C , and -25°C , respectively.

characteristics of the explored CFET follow our earlier work [33]. Figs. 9(a)-(b) illustrate I_{DS} - V_{DS} fitting results across different temperatures and V_{GS} values for both P- and N-FET of CFETs, with RMSE values below 0.16%, where the blue circles correspond to simulated data and the black and red lines represent ANN model's results for $T = -25^\circ\text{C}$, 25°C , and 75°C . Figs. 9(c)-(d) show the I_{DS} - V_{GS} fitting results for the P- and N-FET of CFETs at $V_{DS} = 0.8$ and 0.05 V, respectively. The ANN model achieves a fitting RMSE below 0.23%. Additionally, the C_{GS} - V_{GS} and C_{GD} - V_{GS} curves, as shown in Figs. 9(e)-(h), demonstrate RMSE values below 0.2%. These results confirm that the proposed physics-based ANN compact model not only provides accurate predictions of I - V and C - V with respect

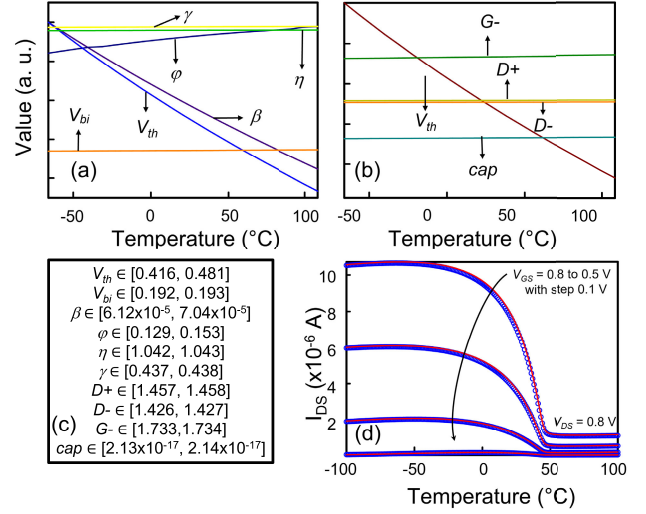


FIGURE 8. Temperature-dependent trends of (a) five ANN1 outputs and V_{th} used in the GF model and (b) four ANN1 outputs and V_{th} used in the MY model, with T varying from -50°C to 100°C . The device is with $L_G = 38$ nm and $T_{FIN} = H_{FIN} = 21$ nm. (c) Numeric ranges of ANN1 outputs. (d) Predicted I_{DS} as a function of temperature for V_{GS} from 0.5 to 0.8 V and $V_{DS} = 0.8$ V of our model.

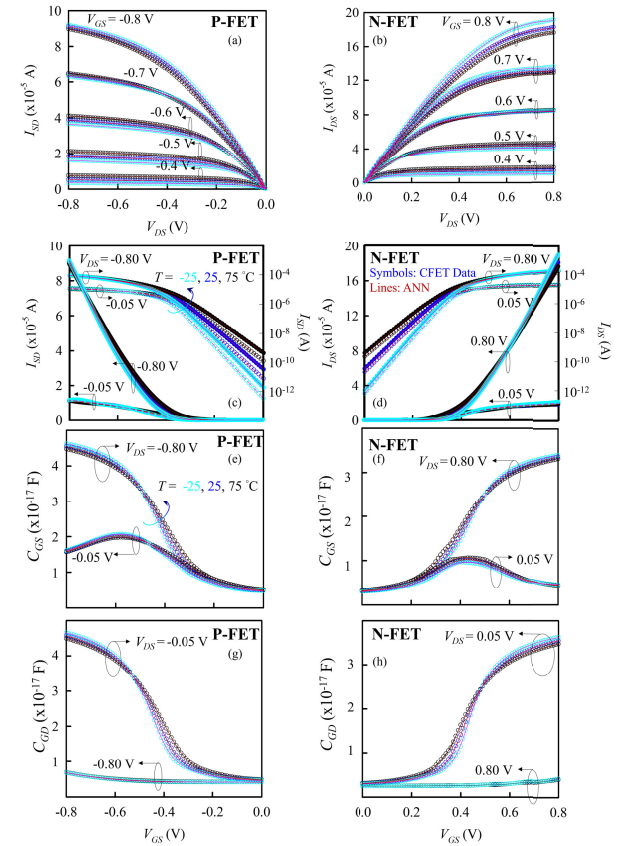
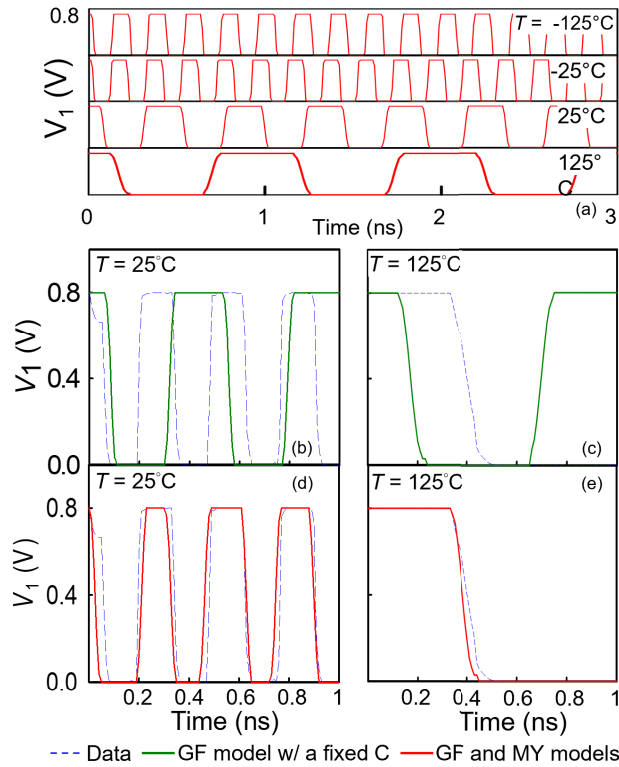


FIGURE 9. Physics-informed ANN model results of (a) (c) (e) (g) P- and (b) (d) (f) (h) N-FET of CFET at different temperatures.

to T for the explored GAA NS-FETs, but also achieves high accuracy and fast convergence (see Fig. 2(b)) for CFET

TABLE 3. Estimated averaged power consumption of logic circuits.

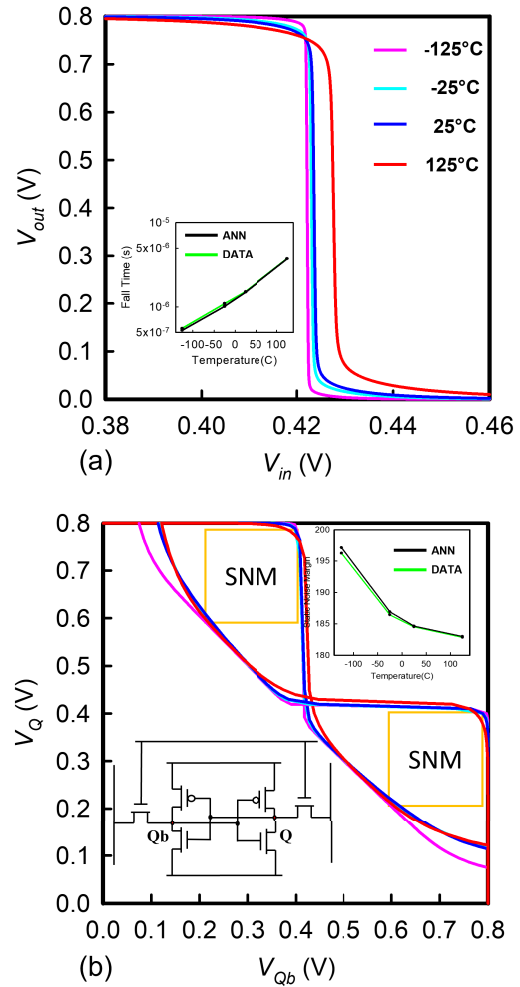
CMOS Logics	Inverter	AND	NAND	OR	NOR
P (pW)	26.7	97.9	74.7	73.8	87.6
CMOS Logics	XOR	XNOR	FA	DFF	11-RO
P (pW)	79.3	132.7	203.7	177.3	5.6×10^6

**FIGURE 10.** (a) The waveform of 11-stage RO with MY model in different T . Accuracy comparison of the timing of 11-stage RO between the model (b)-(c) with a constant capacitance (green) and (d)-(e) MY model (red) for $T = 25^\circ\text{C}$ and 125°C .

simulation data, affirming its versatility and robustness for advanced device modeling.

B. ANN MODELS FOR MULTI-TEMPERATURE CIRCUIT SIMULATION

To evaluate the proposed ANN models in circuit-level applications, we simulate an 11-stage ring oscillator (RO), an inverter, and a 6T SRAM as shown in Figs. 10 and 11. As T increases, the oscillation frequency of RO (Fig. 10(a)) decreases due to a degraded subthreshold swing [34]. The work reported in [25] has mentioned the temperature effect, but the modeling results were not presented. The effect of intrinsic device capacitance is verified using the proposed models; Figs. 10(b)-(c) and (d)-(e) compare the model with a fixed value of gate capacitance $C_{GS,N-FET} = 3.3 \times 10^{-17}$ F and $C_{GS,P-FET} = 4.5 \times 10^{-17}$ F against the constructed C - V model at 25°C and 125°C . The green lines representing the GF model with (w/) a fixed capacitance show clear discrepancies from the data (blue). In contrast,

**FIGURE 11.** The simulated (a) VTC of inverter and (b) butterfly curves of 6T SRAM based on GAA NS-FET using the physics-informed ANN models at different temperatures.

the red lines are the results from the GF and MY models, simultaneously, reducing the error to below 0.57%, demonstrating the proposed ANN models' ability to accurately capture waveforms in high-speed circuits. Fig. 10 validates the trained ANN models through simulations of CMOS circuits based on 26-nm NS-FETs. In Fig. 11(a), the voltage transfer characteristics (VTC) of an inverter are shown across various temperatures. As temperature decreases, the fall time (t_f) becomes shorter, indicating a sharper transition, while higher temperatures result in a more gradual slope. Fig. 11(b) presents the butterfly curves of a 6T SRAM cell, a standard metric for evaluating memory stability. The extracted static noise margin (SNM) values at -125°C , -25°C , 25°C , and 125°C are 196.81, 187.11, 184.52, and 182.85 mV, respectively. For comparison, the corresponding values from BSIM-CMG simulations are 197.11, 186.91, 184.64, and 182.99 mV. The calculated RMSE of SNM between the ANN predictions and BSIM-CMG data remains below

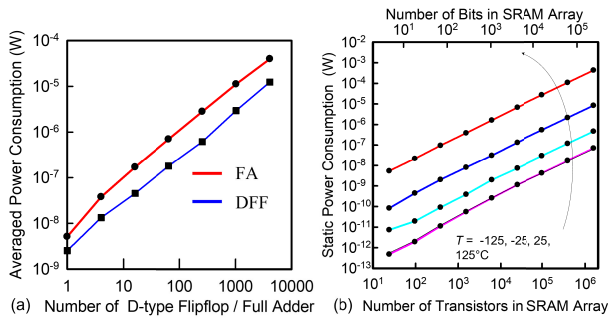


FIGURE 12. (a) The averaged total power consumption vs. the number of DFF and FA. (b) The static power consumption vs. the number of transistors and bits in 6T SRAM array.

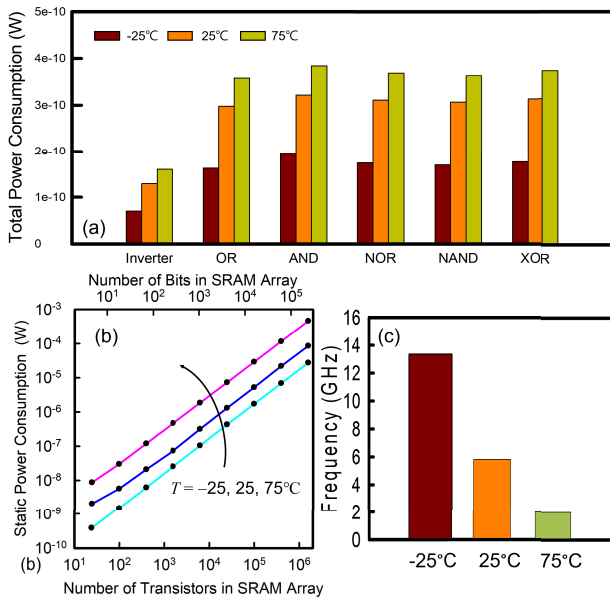


FIGURE 13. The (a) averaged power consumption of the logic gates, (b) static power consumption vs. the number of transistors and bits in 6T SRAM array, and (c) oscillation frequency of the 11-stage RO of CFET models in -25°C , 25°C and 75°C .

TABLE 4. Comparison of time cost between the ANN model and BSIM-CMG in a commercial circuit simulator.

Circuits	CPU time (ms)	
	This work	BSIM-CMG
CMOS inverter	140	370
OR gate	160	450
11-stage RO	950	4180

0.5% across all temperatures. These results indicate that lower temperatures enhance circuit stability, whereas higher temperatures degrade it. Collectively, they confirm that the studied ANN models accurately capture T -induced variations in circuit behavior, while maintaining good agreement with the results of BSIM-CMG. We apply the proposed ANN models to examine the power consumption of various CMOS circuits of GAA NS-FETs.

Table 3 summarizes the averaged power consumption (P) for the examined CMOS logic gates and sequential

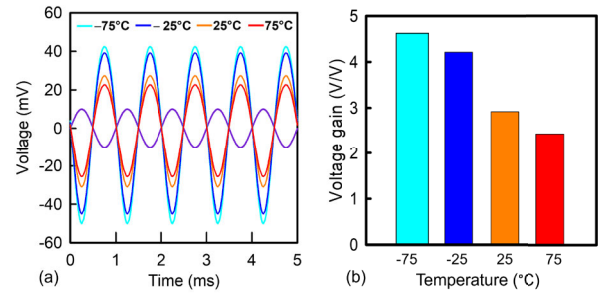


FIGURE 14. Simulation results of the CS amplifier based on CFET devices at various temperatures. (a) Transient output waveforms under sinusoidal input. (b) Extracted voltage gain at temperatures ranging from -75°C to 75°C , showing a clear reduction in gain with increasing temperature.

circuits at $T = 25^\circ\text{C}$, which includes the dynamic, short-circuit, and static powers. The magnitude of P varies from 26.7 (Inverter) to 132.7 (XOR) pW. It increases as the complexity of logic circuit grows. Among the examined logic circuits, the full adder (FA) and D-type flip-flop (DFF) consume 203.7 and 177.3 pW, respectively, reflecting their intricate internal structures and the additional switching activities required during operation. Notably, the 11-stage ring oscillator (RO) consumes the highest amount of P , reaching $5.6 \mu\text{W}$, due to repeated switching events inherent in sequential logic circuits. Fig. 12(a) presents the average total power consumption of combinational and sequential logic circuits, including FA and DFF, at room temperature. As the number of components increases, power consumption rises proportionally. Fig. 12(b) shows the static power consumption versus the number of transistors in the 6T SRAM array across different temperatures. The results indicate that as T increases, the static power consumption also rises significantly due to elevated leakage currents. Moreover, the impact of temperature on power consumption is substantially greater than that of the number of transistors. Our ANN models can reasonably capture these thermally induced trends and reliably predicts power variations in large-scale and complex circuit architectures. The capability of our ANN models is particularly critical for the design of memory and high-density logic circuits operating across a wide range of thermal conditions.

Figs. 13(a)-(c) show circuit-level simulation results based on the constructed ANN CFET models for $T = -25^\circ\text{C}$, 25°C , and 75°C . As shown in Fig. 13(a), the averaged power consumption of logic gates increases with temperature, while Fig. 13(b) confirms that the static power consumption of the SRAM array scales with both the number of transistors and temperature. This behavior aligns with the physical trends observed in Fig. 12. Furthermore, both figures demonstrate the capability of our physics-informed ANN model to simulate chip-level circuits with over 1.5 million transistors, confirming its scalability and practical applicability in large-scale VLSI design. Fig. 13(c) further illustrates the oscillation frequency of an 11-stage RO under different thermal conditions. The frequency decreases as temperature

TABLE 5. Summary of recent ANN works for transistor characteristic modeling.

Models	[28]	[18]	[19]	[25]	This work
Implemented devices	GAA FETs, CFET	GAA CSFET	GAA FETs	GAA FETs	GAA FETs, CFET
Model size	ANN1: (10, 20, 15) ANN2: (10, 20, 10)	(20, 20)	(8, 8, 8)	Global: (20, 15) Local: (20, 15)	ANN1: (8, 8, 8) ANN2: (8, 8, 8)
Extension to new devices	Yes	Yes	Yes	Yes	Yes
Temperature range	Room temperature	Room temperature	Room temperature	Room temperature	-100°C – 100°C
Inputs	$V_{DS}, V_{GS}, L_G, T_{FIN},$ $H_{FIN}, T_{OXP}, EOT,$ N_{BODY}, N_{SD}	V_{DS}, V_{GS}	V_{DS}, V_{GS}	$V_{DS}, V_{GS}, L_G, W, T$	$V_{DS}, V_{GS}, L_G, T, T_{FIN},$ $H_{FIN}, T_{OXP}, EOT, N_{BODY},$ $N_{SD}, N_{channel}$
Outputs	I_{DS}	I_{DS}, C_{GS}, C_{GD}	I_{DS}, Q_G, Q_D, Q_S	I_{DS}, Q_G, Q_D, Q_S	I_{DS}, C_{GS}, C_{GD}
Physical meaning	Yes	No	No	No	Yes
Predictability	High	Medium	Low	Low	High
Transient accuracy	Low	High	High	High	Very stable
Various circuit simulation	Logic and SRAM	Logic	Logic	Logic	Logic, SRAM, and analog
Device count of VLSI	$\sim 20\text{k}$	NA	NA	NA	$> 1.5\text{M}$
Training data requirement	Low	High	High	High	Low

risers, consistent with the trend shown in Fig. 10(a) for GAA NS-FET-based RO. These results validate that our physics-informed ANN I - V and C - V models effectively capture the intrinsic device characteristics and accurately predict temperature-dependent circuit behavior for both GAA NS-FETs and CFETs across various circuit configurations and thermal conditions. To further assess the generality of our ANN model beyond digital logic and SRAM circuits, we simulated a common-source (CS) amplifier implemented with CFET devices under varying temperature conditions. Analog circuits are particularly sensitive to device-level parameter variations, making them a suitable test case for evaluating the physical accuracy of compact models. As shown in Fig. 14, the model accurately captures the impact of temperature on the amplifier's voltage gain, which decreases as temperature increases from -75°C to 75°C . The successful modeling of temperature-dependent behavior in an analog circuit further demonstrates the robustness and applicability of our physics-informed ANN model across a wide range of circuit types.

In large-scale designs, computational efficiency is a key factor, as complex circuits with a vast number of devices can incur significant simulation time. Table 4 compares the simulation time between our ANN models and the BSIM-CMG2024 model for the same tested circuits. Specifically, for the tested CMOS inverter circuit, our ANN models complete a simulation in 140 ms, which is more than 60% faster than that of the BSIM-CMG (370 ms). For the OR gate circuit, the ANN models take 160 ms, compared to 450 ms using the BSIM-CMG. The improvement of the computational cost becomes even more pronounced for larger circuits, such as the 11-stage RO, where the simulation involves a time step of 10 ps over a total duration of 30 ns. The ANN models complete the task in 950 ms, whereas the BSIM-CMG requires 4180 ms, indicating that the ANN models are more than four times faster. This reduction in execution time cost (improved by a factor of 4–4.5 $\times$) of circuit simulation highlights the ANN models' ability to handle complex,

high-frequency circuits more efficiently, making it a highly valuable tool for advanced circuit simulations. We also verify the memory usage for the aforementioned circuits; our peak memory usage is lower than that of the BSIM-CMG, and a maximum difference reaches 10% is observed.

C. BENCHMARK COMPARISON AMONG ANN MODELS

Finally, we compare our work against recently reported ANN models for device characteristic modeling, as listed in Table 5. The comparison considers key aspects including prediction accuracy of device characteristics: I - V and C - V curves, physical interpretability, and accuracy in dynamic circuit simulation (referred to as transient accuracy, which reflects the fidelity of circuit-level transient responses governed by the underlying I - V and C - V models), etc. Our ANN models consistently outperform traditional ANN-based models in various aspects. For example, while most models [18], [19], [25], [28] focus on a limited set of devices or input parameters, our model generalizes across both GAA NS-FETs and CFETs while incorporating a broader range of critical input parameters. This enables accurate prediction of both current (I_{DS}) and capacitances (C_{GS} and C_{GD}) that many prior models lack, especially when accurately modeling circuit behavior, such as transfer and output characteristics, timing, and dynamic properties. In addition, most existing models are designed for fixed or narrow temperature ranges, whereas our model is widely validated from -100°C to 100°C , making it suitable for practice applications under varying environmental conditions. Moreover, our ANN models stand out due to its strong physical interpretability, setting it apart from purely data-driven models that lack this aspect. The incorporation of physical laws not only enhances the accuracy of predictions but also allows our model to generalize better across different devices and operational conditions, even outside of the training region (at least 25% extrapolation capability). Furthermore, our ANN models' reliance on both ML and physical principles of FETs significantly reduces the training data requirements,

TABLE 6. List of symbols and their physical meanings used in this work.

Symbol	Category	Physical Meaning
T	Thermal	Device operating temperature
t_{rel}	Thermal	Relative temperature
T_{ref}	Thermal	Reference temperature
L_G	Device parameter	Gate length of the device
T_{FIN}	Device parameter	Fin thickness
H_{FIN}	Device parameter	Fin height
T_{OX}	Device parameter	Physical oxide thickness
EOT	Device parameter	Equivalent oxide thickness
N_{BODY}	Device parameter	Body doping concentration
N_{SD}	Device parameter	Source/drain doping concentration
$N_{channel}$	Device parameter	Number of channels in GAA device
V_{GS}	Bias	Gate-to-source voltage
V_{DS}	Bias	Drain-to-source voltage
I_{DS}	Output	Drain current
C_{GS}	Output	Gate-to-source capacitance
C_{GD}	Output	Gate-to-drain capacitance
V_{bi}	GF factor	Built-in potential related to device electrostatics
η	GF factor	Parameter accounting for short-channel or field effects
β	GF factor	Mobility-related parameter in current formulation
γ	GF factor	Body effect coefficient affecting threshold voltage
φ	GF factor	Surface potential representing channel electrostatics
V_{th}	GF and MY factor	Threshold voltage
cap	MY factor	Gate oxide capacitance over effective channel area
G^-	MY factor	Smoothing factor
D^-	MY factor	Smoothing factor
D^+	MY factor	Smoothing factor
x	ANN input	Input vector
z	ANN1 output	GF/MY model parameters predicted by ANN1
o	ANN2 output	Log-scale correction factors for $I-V$ and $C-V$ predicted by ANN2
ε_1	Correction	I_{DS} correction factor
ε_2	Correction	C_{GD} correction factor
ε_3	Correction	C_{GS} correction factor
y	Target	Ground-truth vector
$\hat{y}$	Prediction	Model-predicted values
θ_1	ANN	Trainable parameters of ANN1
θ_2	ANN	Trainable parameters of ANN2

which is a major advantage over other approaches that often require large datasets to achieve engineering reasonable results inside the range of model. The combination of high accuracy, physical meaning, broad applicability, and reduced data requirements makes our ANN models a promising advancement in compact modeling of emerging FETs, serving as a robust approach for circuit simulations across a wide range of device dimensions, bias conditions, and operational temperatures.

IV. CONCLUSION

We have presented a device-physics-informed ANN framework for thermal-aware $I-V$ and $C-V$ modeling of GAA

FETs. By embedding the Grove–Frohm and Meyer models within a dual-ANN structure, our approach accurately captures temperature-dependent device behavior while maintaining physical interpretability. The model achieves RMSE below 0.6% across a wide range of voltages, device dimensions, and temperatures, including conditions beyond the training data. Notably, it enables fast convergence and reduced training data and is, to our knowledge, the first ANN-based model successfully applied to simulate a 1.5-million-transistor SRAM circuit, demonstrating both scalability and practical utility. Future work will extend the framework to non-silicon devices, including III–V semiconductors and 2D materials, and explore its application in extracting parasitic resistances and capacitances across diverse circuit layouts.

APPENDIX

SUMMARY OF SYMBOLS AND PHYSICAL MEANINGS

This appendix summarizes all key symbols used throughout this work, along with their corresponding physical meanings and functional roles in the proposed modeling framework. The symbols are grouped into categories including thermal conditions, device parameters, electrical biases, outputs, GF/MY model parameters, and neural network variables.

For clarity, the presented notation is consistent with both conventional device physics and the hybrid modeling methodology adopted in this paper, which integrates physics-based formulations (GF/MY models) with data-driven neural network corrections.

REFERENCES

- [1] K. Mehta and H.-Y. Wong, “Prediction of FinFET current-voltage and capacitance-voltage curves using machine learning with autoencoder,” *IEEE Electron Device Lett.*, vol. 42, no. 2, pp. 136–139, Feb. 2021.
- [2] H. M. Koçak, J. Davis, M. Houssa, A. T. Naskali, and J. Mitard, “Machine learning-based universal threshold voltage extraction of transistors using convolutional neural networks,” *IEEE Trans. Semicond. Manuf.*, vol. 37, no. 4, pp. 615–619, Nov. 2024.
- [3] J. Ha, S. J. Kim, M. Bang, G. Lee, M. Suh, M. Shim, C.-E. Kim, and J. Kim, “Prediction of statistical distribution on nanosheet FET by geometrical variability using various machine learning models,” *IEEE Access*, vol. 11, pp. 125217–125225, 2023.
- [4] K. Ko, J. K. Lee, M. Kang, J. Jeon, and H. Shin, “Prediction of process variation effect for ultrascaled GAA vertical FET devices using a machine learning approach,” *IEEE Trans. Electron Devices*, vol. 66, no. 10, pp. 4474–4477, Oct. 2019.
- [5] C. Akbar, Y. Li, and W.-L. Sung, “Deep learning algorithms for the work function fluctuation of random nanosized metal grains on gate-all-around silicon nanowire MOSFETs,” *IEEE Access*, vol. 9, pp. 73467–73481, 2021.
- [6] P. Jungmann, J. B. Johnson, E. C. Silva, W. Taylor, A. H. Khan, and A. Kumar, “TCAD-enabled machine learning—An efficient framework to build highly accurate and reliable models for semiconductor technology development and fabrication,” *IEEE Trans. Semicond. Manuf.*, vol. 36, no. 2, pp. 268–278, May 2023.
- [7] C.-C. Chiang, H.-Y. Lan, L. Liu, Y. P. Chen, D. Zemlyanov, J. Appenzeller, and Z. Chen, “Design and process co-optimization of 2-D monolayer transistors via machine learning,” *IEEE Trans. Electron Devices*, vol. 70, no. 11, pp. 5991–5996, Nov. 2023.
- [8] J. P. Duarte, S. Khandelwal, A. Medury, C. Hu, P. Kushwaha, H. Agarwal, A. Dasgupta, and Y. S. Chauhan, “BSIM-CMG: Standard FinFET compact model for advanced circuit design,” in *Proc. ESSCIRC Conf.-41st Eur. Solid-State Circuits Conf. (ESSCIRC)*. Cincinnati, OH, USA: Standard, Sep. 2015, pp. 196–201.

- [9] Y. Li, H.-M. Chou, and J.-W. Lee, "Investigation of electrical characteristics on surrounding-gate and omega-shaped-gate nanowire FinFETs," *IEEE Trans. Nanotechnol.*, vol. 4, no. 5, pp. 510–516, Sep. 2005.
- [10] Y. Li, H.-W. Cheng, Y.-Y. Chiu, C.-Y. Yiu, and H.-W. Su, "A unified 3D device simulation of random dopant, interface trap and work function fluctuations on high- κ /metal gate device," in *IEDM Tech. Dig.*, Dec. 2011, pp. 1–4.
- [11] Y. Li, H.-T. Chang, C.-N. Lai, P.-J. Chao, and C.-Y. Chen, "Process variation effect, metal-gate work-function fluctuation and random dopant fluctuation of 10-nm gate-all-around silicon nanowire MOSFET devices," in *IEDM Tech. Dig.*, Dec. 2015, pp. 1–4.
- [12] A. Razavi, P. Zeitoff, and E. J. Nowak, "Challenges and limitations of CMOS scaling for FinFET and beyond architectures," *IEEE Trans. Nanotechnol.*, vol. 18, pp. 999–1004, 2019.
- [13] M.-H. Chuang, S. R. Kola, and Y. Li, "Impact of bottom channel coverage ratio on electrical characteristics of GAA Si NS CFETs for sub-1-nm nodes," *Solid-State Electron.*, vol. 230, Dec. 2025, Art. no. 109244.
- [14] K. Sheelvardhan, S. Guglani, M. Ehteshamuddin, S. Roy, and A. Dasgupta, "Machine learning augmented compact modeling for simultaneous improvement in computational speed and accuracy," *IEEE Trans. Electron Devices*, vol. 71, no. 1, pp. 239–245, Jan. 2024.
- [15] P. Raut, D. K. Panda, and A. K. Goyal, "A comprehensive review on next-generation modeling and optimization for semiconductor devices," *IEEE Access*, vol. 13, pp. 123724–123742, 2025.
- [16] K. M. Mamun, N. Pala, and M. S. A. Shawkat, "A comprehensive review of machine learning approaches for semiconductor device modeling and simulation," *IEEE Access*, vol. 13, pp. 162969–162991, 2025.
- [17] X. Li, Z. Wu, G. Rzepa, M. Karner, H. Xu, Z. Wu, W. Wang, G. Yang, Q. Luo, L. Wang, and L. Li, "Overview of emerging semiconductor device model methodologies: From device physics to machine learning engines," *Fundam. Res.*, vol. 5, no. 5, pp. 2149–2160, 2024.
- [18] Q. Yang, G. Qi, W. Gan, Z. Wu, H. Yin, T. Chen, G. Hu, J. Wan, S. Yu, and Y. Lu, "Transistor compact model based on multigradient neural network and its application in SPICE circuit simulations for gate-all-around silicon source FETs," *IEEE Trans. Electron Devices*, vol. 68, no. 9, pp. 4181–4188, Sep. 2021.
- [19] C.-T. Tung, M.-Y. Kao, and C. Hu, "Neural network-based modeling with high accuracy and potential model speed," *IEEE Trans. Electron Devices*, vol. 69, no. 11, pp. 6476–6479, Nov. 2022.
- [20] C.-T. Tung, A. Pampori, C. Kumar Dabhi, S. Salahuddin, and C. Hu, "A novel neural network-based transistor compact model including self-heating," *IEEE Electron Device Lett.*, vol. 45, no. 8, pp. 1512–1515, Aug. 2024.
- [21] M. Li, O. Irsoy, C. Cardie, and H. G. Xing, "Physics-inspired neural networks for efficient device compact modeling," *IEEE J. Explor. Solid-State Comput. Devices Circuits*, vol. 2, pp. 44–49, 2016.
- [22] S. M. Sze, Y. Li, and K. K. Ng, *Physics of Semiconductor Devices*, 4th ed., Hoboken, NJ, USA: Wiley, 2021.
- [23] M. Zhu, L. Wang, H. Li, C. Yang, D. Ma, and F. Yang, "Characteristics of SiC MOSFET in a wide temperature range," in *Proc. IEEE Workshop Wide Bandgap Power Devices Appl. Asia (WiPDA Asia)*, Aug. 2021, pp. 79–82.
- [24] M. H. Bhuyan, F. Ferdous, and Q. D. M. Khosru, "Temperature effects on subthreshold current of pocket implanted nano scale n-MOSFET," in *Proc. 7th Int. Conf. Electr. Comput. Eng.*, Dec. 2012, pp. 686–689.
- [25] J. Wang, Y.-H. Kim, J. Ryu, C. Jeong, W. Choi, and D. Kim, "Artificial neural network-based compact modeling methodology for advanced transistors," *IEEE Trans. Electron Devices*, vol. 68, no. 3, pp. 1318–1325, Mar. 2021.
- [26] D. Frohman-Bentchkowsky and A. Grove, "Conductance of MOS transistors in saturation," *IEEE Trans. Electron Devices*, vol. ED-16, no. 1, pp. 108–113, Jan. 1969.
- [27] J. E. Meyer, "MOS models and circuit simulation," *RCA Rev.*, vol. 32, pp. 42–63, Mar. 1971.
- [28] Y.-S. Yang, Y. Li, and S. R. Kola, "A physical-based artificial neural networks compact modeling framework for emerging FETs," *IEEE Trans. Electron Devices*, vol. 71, no. 1, pp. 223–230, Jan. 2024.
- [29] (2024). *International Roadmap for Devices and Systems (IRDS) 2024 Metrology*. [Online]. Available: <https://irds.ieee.org/images/files/pdf/2024/2024IRDSMET.pdf>
- [30] S. Dash and Y. Li, "Transfer learning approach to modeling multichannel gate-all-around nanosheet field-effect transistors under work function fluctuations," *Eng. Appl. Artif. Intell.*, vol. 148, May 2025, Art. no. 110322.
- [31] C. C. McAndrew, "Validation of MOSFET model source-drain symmetry," *IEEE Trans. Electron Devices*, vol. 53, no. 9, pp. 2202–2206, Sep. 2006.
- [32] Y. Dei, Y.-S. Yang, and Y. Li, "Compact modeling of N- and P-type GAA NS FETs using physical-based artificial neural networks with temperature dependence," in *Proc. Int. Conf. Simul. Semiconductor Processes Devices (SISPAD)*, Sep. 2023, pp. 285–288.
- [33] S. R. Kola and Y. Li, "Electrical characteristic and power fluctuations of GAA Si NS CFETs by simultaneously considering six process variation factors," *IEEE Open J. Nanotechnol.*, vol. 4, pp. 229–238, 2023.
- [34] Y. Wang, P. K. Chan, and K. H. Li, "A compact CMOS ring oscillator with temperature and supply compensation for sensor applications," in *Proc. IEEE Comput. Soc. Annu. Symp. VLSI*, Jul. 2014, pp. 267–272.



YUN TAI received the B.S. degree from the Department of Electrical and Computer Engineering, National Chiao Tung University, Hsinchu, Taiwan, in 2019, and the M.S. degree from the Parallel and Scientific Computing Laboratory, Institute of Biomedical Engineering, National Yang Ming Chiao Tung University, Hsinchu, in 2025. His research interest includes machine learning for device modeling.



YIMING LI (Senior Member, IEEE) has been a Visiting Professor with Grenoble INP, Stanford University, and Tohoku University, since 2011. He is currently a Full Professor of electronics and electrical engineering with National Yang Ming Chiao Tung University, Hsinchu, Taiwan. He is the author or co-author of more than 400 technical articles, including the book *Physics of Semiconductor Devices* (4th Edition, Wiley), in 2021. His current research interests include computational electronics, device physics, semiconductor nanostructures, modeling and parameter extraction, biomedical and energy harvesting devices, and optimization techniques. He has been a Program Committee Member of IEDM, since 2011.



MIN-HUI CHUANG (Student Member, IEEE) received the B.S. degree from the Department of Electrical and Computer Engineering, National Chiao Tung University, Hsinchu, Taiwan, in 2019, and the M.S. degree from the Institute of Communications Engineering, National Chiao Tung University, in 2020. She is currently pursuing the Ph.D. degree with the Parallel and Scientific Computing Laboratory, National Yang Ming Chiao Tung University, Hsinchu. Her research interests include numerical semiconductor device simulation, compact model parameter extraction, and machine learning of device modeling.

...